

Titration report



Method

Method ID	CDC_M003	Release state	Released
Name	Titration with HCl - Liquid	Modified by	PredefinedUser
Modified on	2/25/2026 4:01:53 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.010940 mL
		Concentration R2	0.10022 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.022725 mL
		Concentration R2	0.10045 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.012918 mL
		Concentration R2	0.10026 mol/L

Statistics

R1

Mean	5.015528 mL	Minimum	5.010940 mL
Standard deviation	0.006311 mL	Rel. standard deviation	0.126 %
Total samples	3	Samples excluded	0

R2

Mean	0.10031 mol/L	Minimum	0.10022 mol/L
Standard deviation	0.00012 mol/L	Rel. standard deviation	0.123 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165279	Task name	Unused DG111 - INC = 0.1 mL
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.010940 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10022 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.022725 mL	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10045 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.012918 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10026 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:03:28 PM
Overall state	OK
Sample start time	2/25/2026 4:03:28 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:04:04 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:04:04 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	2/25/2026 4:04:32 PM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 4:06:32 PM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	99.7 mV
EHNV1	-72.5 mV
Q1	0.501094 mmol
VEQ1	5.010940 mL
EST	-177.9 mV
VEX	0.889060 mL
VEND	5.9000 mL
t	03:25 min
QEND	0.59000 mmol
QEX	0.088906 mmol
nEQ	1

Consumption (R1)

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Creation time	2/25/2026 4:10:02 PM
Overall state	OK
Result	5.010940 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 4:10:02 PM
Overall state	OK
Result	0.10022 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:10:22 PM
Overall state	OK
Sample start time	2/25/2026 4:10:22 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:10:24 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:10:24 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:13 min

Stir (Stir1)

Creation time	2/25/2026 4:10:51 PM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 4:12:52 PM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	98.3 mV
EHNv1	-73.0 mV
Q1	0.502272 mmol
VEQ1	5.022725 mL

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EST	-176.6 mV
VEX	0.977275 mL
VEND	6.0000 mL
t	03:27 min
QEND	0.60000 mmol
QEX	0.097728 mmol
nEQ	1

Consumption (R1)

Creation time	2/25/2026 4:16:24 PM
Overall state	OK
Result	5.022725 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 4:16:24 PM
Overall state	OK
Result	0.10045 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:16:46 PM
Overall state	OK
Sample start time	2/25/2026 4:16:46 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:16:48 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:16:48 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	2/25/2026 4:17:15 PM
Overall state	OK

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Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 4:19:16 PM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	99.7 mV
EHN1	-71.5 mV
Q1	0.501292 mmol
VEQ1	5.012918 mL
EST	-176.5 mV
VEX	0.987082 mL
VEND	6.0000 mL
t	03:23 min
QEND	0.60000 mmol
QEX	0.098708 mmol
nEQ	1

Consumption (R1)

Creation time	2/25/2026 4:22:43 PM
Overall state	OK
Result	5.012918 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

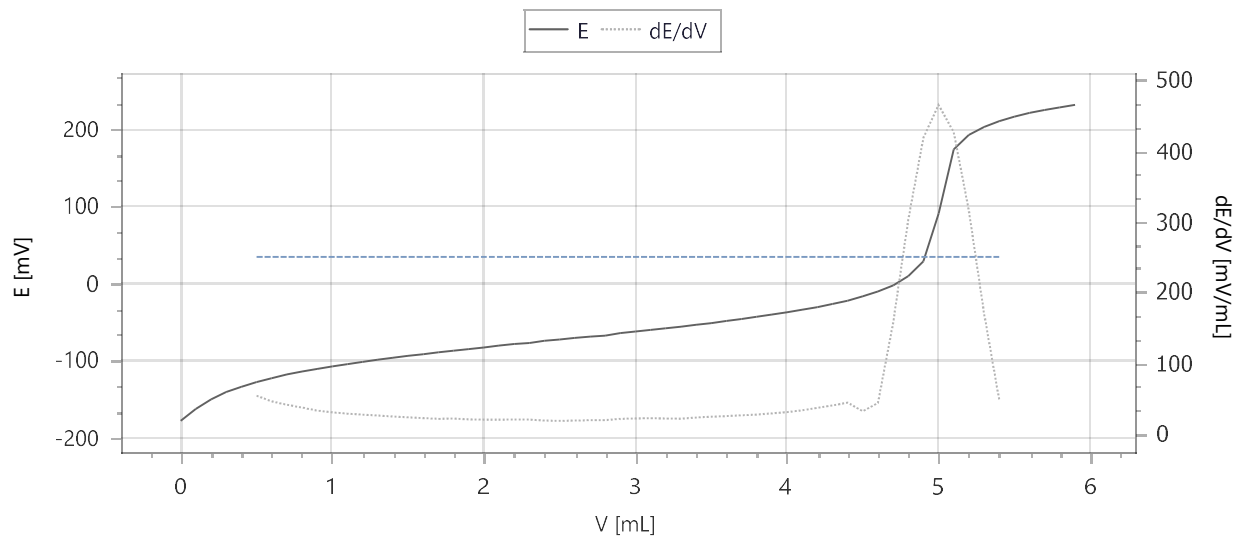
Creation time	2/25/2026 4:22:44 PM
Overall state	OK
Result	0.10026 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-177.9		25.0			
3	0.1000	-162.7	0.1000	25.0	15.2		
6	0.2000	-150.2	0.1000	25.0	12.5		
9	0.3000	-140.5	0.1000	25.0	9.7		
12	0.4000	-134.0	0.1000	25.0	6.5		
15	0.5000	-127.9	0.1000	25.0	6.1	54.38	
18	0.6000	-122.7	0.1000	25.0	5.2	46.41	
21	0.7000	-117.9	0.1000	25.0	4.8	41.54	
24	0.8000	-114.1	0.1000	25.0	3.8	37.68	
28	0.9000	-110.8	0.1000	25.0	3.3	33.38	
30	1.0000	-107.3	0.1000	25.0	3.5	30.91	
34	1.1000	-104.4	0.1000	25.0	2.9	29.03	
37	1.2000	-101.6	0.1000	25.0	2.8	27.69	
40	1.3000	-98.9	0.1000	25.0	2.7	26.47	
43	1.4000	-96.1	0.1000	25.0	2.8	25.00	
46	1.5000	-93.9	0.1000	25.0	2.2	23.90	
49	1.6000	-91.5	0.1000	25.0	2.4	22.75	
52	1.7000	-89.1	0.1000	25.0	2.4	21.79	
55	1.8000	-87.2	0.1000	25.0	1.9	22.08	
58	1.9000	-85.0	0.1000	25.0	2.2	20.99	
61	2.0000	-82.8	0.1000	25.0	2.2	20.57	
64	2.1000	-80.7	0.1000	25.0	2.1	20.53	
68	2.2000	-78.4	0.1000	25.0	2.3	20.73	
71	2.3000	-77.4	0.1000	25.0	1.0	20.65	
74	2.4000	-74.3	0.1000	25.0	3.1	19.31	
77	2.5000	-72.6	0.1000	25.0	1.7	18.96	
80	2.6000	-70.6	0.1000	25.0	2.0	19.31	
83	2.7000	-68.9	0.1000	25.0	1.7	19.82	
86	2.8000	-67.6	0.1000	25.0	1.3	19.87	
89	2.9000	-64.5	0.1000	25.0	3.1	21.75	
92	3.0000	-62.4	0.1000	25.0	2.1	22.31	

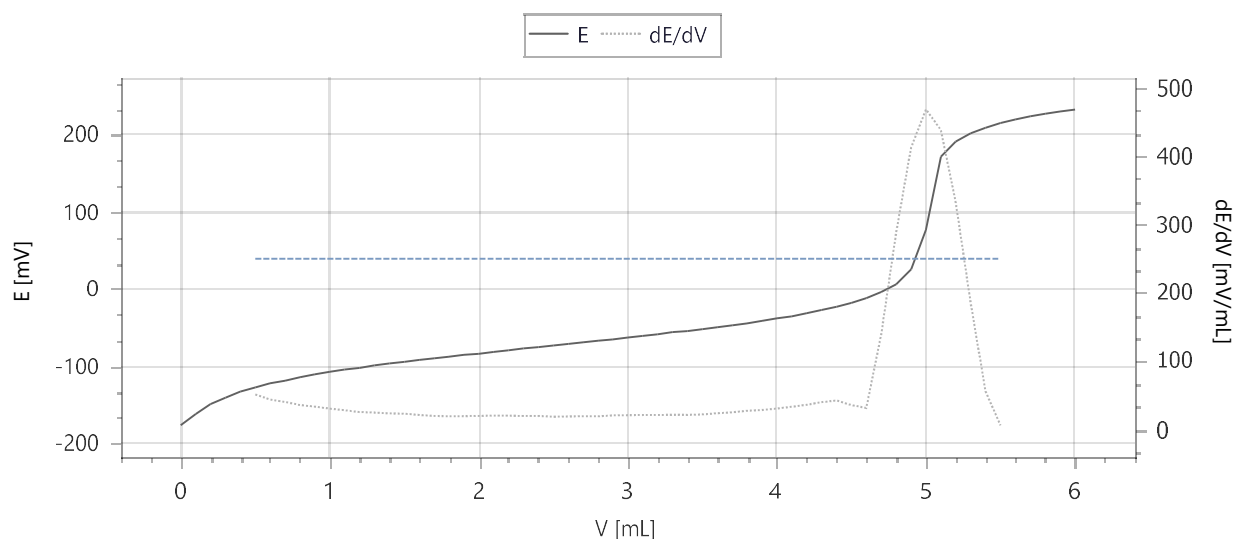
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95	3.1000	-60.4	0.1000	25.0	2.0	22.53	
98	3.2000	-58.1	0.1000	25.0	2.3	22.24	
101	3.3000	-55.9	0.1000	25.0	2.2	21.87	
104	3.4000	-53.7	0.1000	25.0	2.2	23.59	
108	3.5000	-51.3	0.1000	25.0	2.4	24.76	
111	3.6000	-48.7	0.1000	25.0	2.6	25.60	
114	3.7000	-46.1	0.1000	25.0	2.6	26.72	
117	3.8000	-43.2	0.1000	25.0	2.9	27.80	
120	3.9000	-40.5	0.1000	25.0	2.7	29.42	
123	4.0000	-37.6	0.1000	25.0	2.9	31.16	
126	4.1000	-34.2	0.1000	25.0	3.4	33.77	
129	4.2000	-30.6	0.1000	25.0	3.6	37.19	
132	4.3000	-26.5	0.1000	25.0	4.1	40.77	
135	4.4000	-22.3	0.1000	25.0	4.2	44.50	
138	4.5000	-16.8	0.1000	25.0	5.5	32.57	
141	4.6000	-10.5	0.1000	25.0	6.3	44.10	
144	4.7000	-2.6	0.1000	25.0	7.9	156.87	
148	4.8000	9.4	0.1000	25.0	12.0	302.30	
154	4.9000	28.5	0.1000	25.0	19.1	418.20	
157	5.0000	90.6	0.1000	25.0	62.1	464.42	
	5.010940	99.7				464.43	EQP
164	5.1000	173.6	0.1000	25.0	83.0	425.92	
168	5.2000	192.1	0.1000	25.0	18.5	315.33	
171	5.3000	202.7	0.1000	25.0	10.6	170.73	
174	5.4000	210.0	0.1000	25.0	7.3	48.64	
177	5.5000	215.9	0.1000	25.0	5.9		
180	5.6000	220.7	0.1000	25.0	4.8		
184	5.7000	224.6	0.1000	25.0	3.9		
187	5.8000	227.9	0.1000	25.0	3.3		
190	5.9000	231.0	0.1000	25.0	3.1		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



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Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.6		25.0			
3	0.1000	-161.8	0.1000	25.0	14.8		
6	0.2000	-148.9	0.1000	25.0	12.9		
9	0.3000	-140.7	0.1000	25.0	8.2		
12	0.4000	-132.8	0.1000	25.0	7.9		
15	0.5000	-127.4	0.1000	25.0	5.4	52.15	
18	0.6000	-122.2	0.1000	25.0	5.2	44.69	
22	0.7000	-118.8	0.1000	25.0	3.4	41.13	
25	0.8000	-114.2	0.1000	25.0	4.6	36.63	
28	0.9000	-110.5	0.1000	25.0	3.7	34.19	
31	1.0000	-107.1	0.1000	25.0	3.4	31.38	
34	1.1000	-104.3	0.1000	25.0	2.8	28.99	
38	1.2000	-102.0	0.1000	25.0	2.3	26.34	
40	1.3000	-98.9	0.1000	25.0	3.1	25.59	
44	1.4000	-96.4	0.1000	25.0	2.5	24.35	
47	1.5000	-94.1	0.1000	25.0	2.3	24.01	
50	1.6000	-91.7	0.1000	25.0	2.4	22.12	
53	1.7000	-89.6	0.1000	25.0	2.1	20.63	
56	1.8000	-87.7	0.1000	25.0	1.9	20.28	
59	1.9000	-85.2	0.1000	25.0	2.5	20.40	
62	2.0000	-84.0	0.1000	25.0	1.2	20.77	
65	2.1000	-81.3	0.1000	25.0	2.7	21.16	
68	2.2000	-79.3	0.1000	25.0	2.0	21.05	
71	2.3000	-77.0	0.1000	25.0	2.3	20.69	
74	2.4000	-75.1	0.1000	25.0	1.9	20.76	
77	2.5000	-73.2	0.1000	25.0	1.9	19.51	
80	2.6000	-71.2	0.1000	25.0	2.0	19.94	
83	2.7000	-69.1	0.1000	25.0	2.1	20.19	
86	2.8000	-67.0	0.1000	25.0	2.1	20.10	
89	2.9000	-65.2	0.1000	25.0	1.8	21.49	
92	3.0000	-62.8	0.1000	25.0	2.4	21.82	
96	3.1000	-60.8	0.1000	25.0	2.0	22.03	
98	3.2000	-58.7	0.1000	25.0	2.1	22.16	
102	3.3000	-55.5	0.1000	25.0	3.2	22.39	
105	3.4000	-54.4	0.1000	25.0	1.1	22.26	
108	3.5000	-51.8	0.1000	25.0	2.6	22.88	
111	3.6000	-49.4	0.1000	25.0	2.4	24.66	
114	3.7000	-46.8	0.1000	25.0	2.6	25.96	
117	3.8000	-44.3	0.1000	25.0	2.5	28.20	
120	3.9000	-41.2	0.1000	25.0	3.1	29.33	
123	4.0000	-37.8	0.1000	25.0	3.4	31.50	
126	4.1000	-35.4	0.1000	25.0	2.4	34.14	
129	4.2000	-31.3	0.1000	25.0	4.1	36.96	
132	4.3000	-27.2	0.1000	25.0	4.1	40.75	
135	4.4000	-23.0	0.1000	25.0	4.2	43.37	
138	4.5000	-17.7	0.1000	25.0	5.3	36.49	
141	4.6000	-11.8	0.1000	25.0	5.9	32.42	
144	4.7000	-3.8	0.1000	25.0	8.0	139.87	

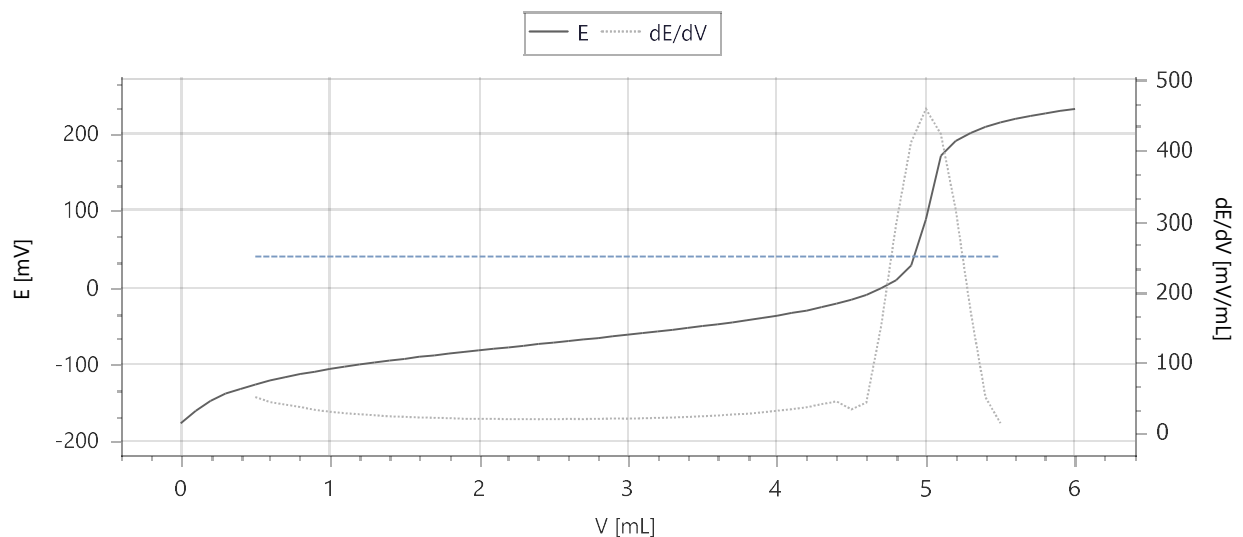
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148	4.8000	6.4	0.1000	25.0	10.2	288.14	EQP
152	4.9000	25.8	0.1000	25.0	19.4	412.08	
155	5.0000	76.7	0.1000	25.0	50.9	467.97	
	5.022725	98.3				468.19	
162	5.1000	171.8	0.1000	25.0	95.1	438.04	
166	5.2000	191.4	0.1000	25.0	19.6	332.31	
170	5.3000	202.1	0.1000	25.0	10.7	185.54	
173	5.4000	209.4	0.1000	25.0	7.3	56.53	
176	5.5000	215.4	0.1000	25.0	6.0	7.02	
179	5.6000	220.1	0.1000	25.0	4.7		
182	5.7000	224.0	0.1000	25.0	3.9		
185	5.8000	227.4	0.1000	25.0	3.4		
188	5.9000	230.3	0.1000	25.0	2.9		
191	6.0000	233.1	0.1000	25.0	2.8		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.5		25.0			
3	0.1000	-160.3	0.1000	25.0	16.2		
6	0.2000	-147.4	0.1000	25.0	12.9		
9	0.3000	-137.8	0.1000	25.0	9.6		
12	0.4000	-131.9	0.1000	25.0	5.9		
15	0.5000	-126.1	0.1000	25.0	5.8	51.49	
18	0.6000	-120.6	0.1000	25.0	5.5	44.33	
21	0.7000	-116.4	0.1000	25.0	4.2	40.65	
24	0.8000	-112.3	0.1000	25.0	4.1	37.38	
28	0.9000	-109.4	0.1000	25.0	2.9	33.25	
30	1.0000	-105.7	0.1000	25.0	3.7	30.56	
34	1.1000	-102.7	0.1000	25.0	3.0	28.53	
37	1.2000	-100.1	0.1000	25.0	2.6	27.00	

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40	1.3000	-97.4	0.1000	25.0	2.7	25.71	
43	1.4000	-95.0	0.1000	25.0	2.4	24.05	
46	1.5000	-92.7	0.1000	25.0	2.3	23.44	
49	1.6000	-90.0	0.1000	25.0	2.7	22.52	
52	1.7000	-88.1	0.1000	25.0	1.9	22.08	
55	1.8000	-85.9	0.1000	25.0	2.2	21.50	
58	1.9000	-83.8	0.1000	25.0	2.1	20.64	
61	2.0000	-81.7	0.1000	25.0	2.1	20.44	
64	2.1000	-79.4	0.1000	25.0	2.3	20.61	
67	2.2000	-77.8	0.1000	25.0	1.6	20.09	
70	2.3000	-75.7	0.1000	25.0	2.1	20.25	
73	2.4000	-73.4	0.1000	25.0	2.3	19.96	
76	2.5000	-71.6	0.1000	25.0	1.8	20.05	
79	2.6000	-69.6	0.1000	25.0	2.0	20.46	
82	2.7000	-67.4	0.1000	25.0	2.2	20.26	
86	2.8000	-65.7	0.1000	25.0	1.7	20.68	
88	2.9000	-63.3	0.1000	25.0	2.4	20.92	
92	3.0000	-61.2	0.1000	25.0	2.1	20.85	
95	3.1000	-59.2	0.1000	25.0	2.0	21.34	
98	3.2000	-57.0	0.1000	25.0	2.2	21.94	
101	3.3000	-55.0	0.1000	25.0	2.0	22.40	EQP
104	3.4000	-52.5	0.1000	25.0	2.5	23.28	
107	3.5000	-50.0	0.1000	25.0	2.5	24.27	
110	3.6000	-47.7	0.1000	25.0	2.3	25.15	
113	3.7000	-45.1	0.1000	25.0	2.6	26.76	
116	3.8000	-42.5	0.1000	25.0	2.6	27.65	
119	3.9000	-39.4	0.1000	25.0	3.1	29.62	
122	4.0000	-36.4	0.1000	25.0	3.0	32.04	
125	4.1000	-32.9	0.1000	25.0	3.5	34.19	
128	4.2000	-29.8	0.1000	25.0	3.1	36.89	
131	4.3000	-25.3	0.1000	25.0	4.5	41.36	
134	4.4000	-20.9	0.1000	25.0	4.4	45.18	
137	4.5000	-15.9	0.1000	25.0	5.0	33.96	
140	4.6000	-9.3	0.1000	25.0	6.6	43.57	
144	4.7000	-0.7	0.1000	25.0	8.6	152.36	
147	4.8000	9.8	0.1000	25.0	10.5	295.90	
150	4.9000	29.1	0.1000	25.0	19.3	411.04	
153	5.0000	89.0	0.1000	25.0	59.9	458.22	
	5.012918	99.7				458.25	
159	5.1000	171.7	0.1000	25.0	82.7	422.95	
163	5.2000	191.0	0.1000	25.0	19.3	315.85	
166	5.3000	201.5	0.1000	25.0	10.5	172.18	
169	5.4000	209.2	0.1000	25.0	7.7	50.92	
172	5.5000	215.0	0.1000	25.0	5.8	14.35	
175	5.6000	219.6	0.1000	25.0	4.6		
178	5.7000	223.5	0.1000	25.0	3.9		
181	5.8000	227.0	0.1000	25.0	3.5		
184	5.9000	230.1	0.1000	25.0	3.1		
187	6.0000	232.6	0.1000	25.0	2.5		

Sample Series

Titration report

LabX

Name

Internal ID

Label 1

Label 2

Label 3

Label 4

Label 5

Label 6

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC = 0.1 mL	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC = 0.1 mL	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:03:26
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC = 0.1 mL	25.02.2026 16:03:26

Used Instruments

Titration report



Name

T9_MSG_SA_002

Serial number

B551866924

Responsible User

Time

Login name

Full name

25.02.2026 16:03:26

PredefinedUser

Administrator